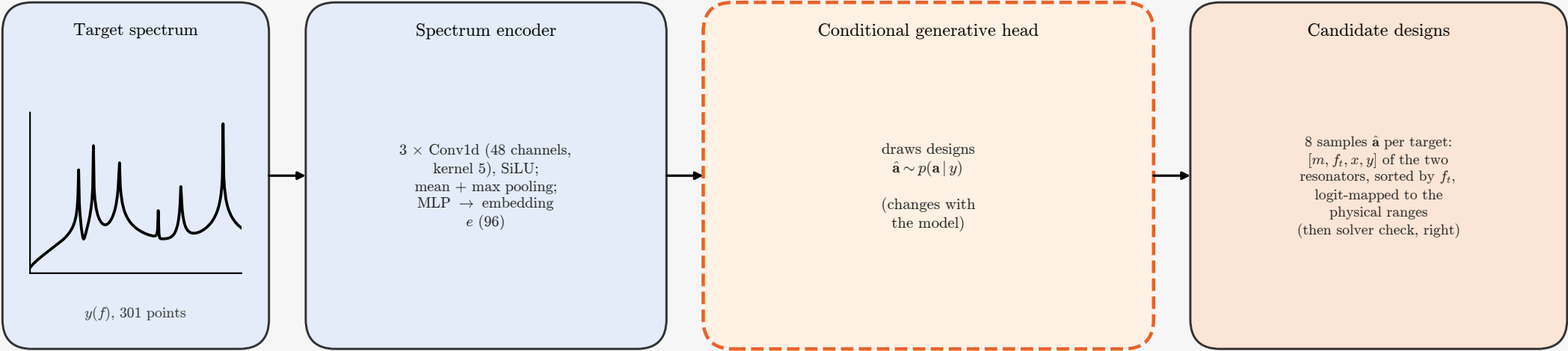


Inverse models: common structure

target ERP spectrum → samples of resonator designs; only the generative head changes between models

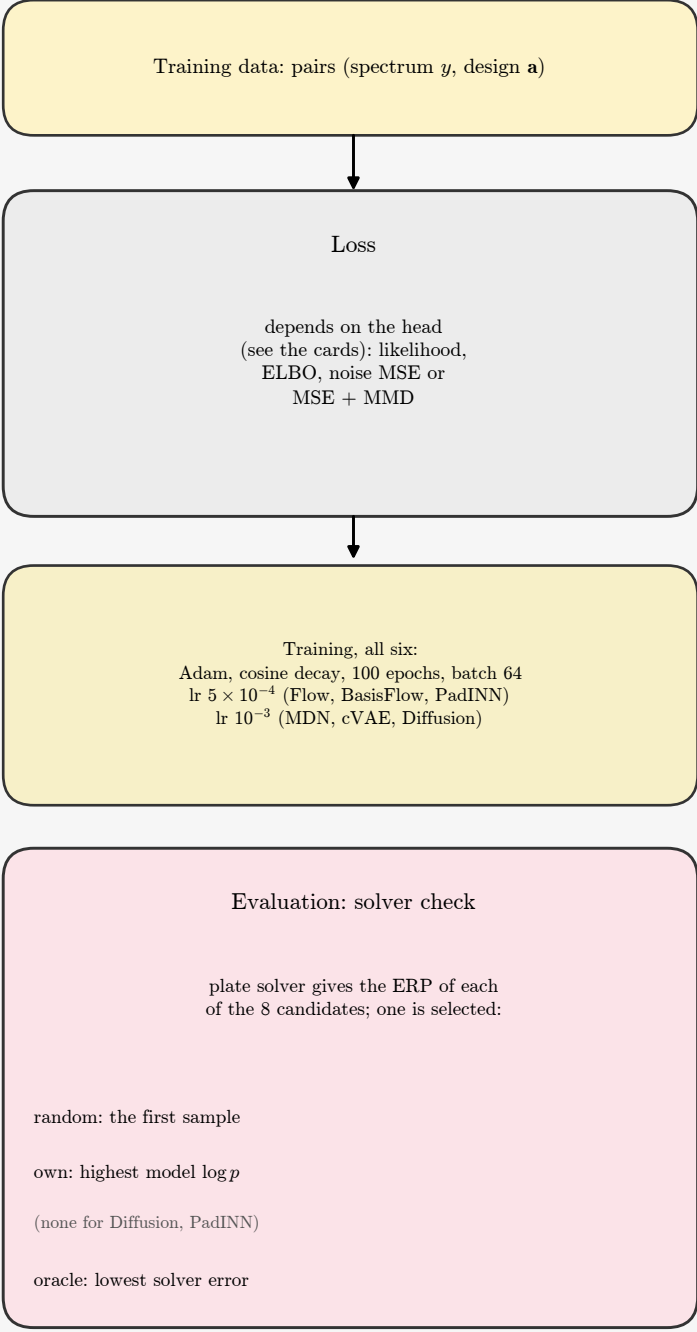


PadINN works on the raw spectrum instead of the encoder embedding; BasisFlow uses the encoder as its learned basis.

Generative head of each model

MDN: mixture density network head: $e \rightarrow$ mixture of 10 diagonal Gaussians over $\mathbf{a}$ (weights, means, log-stds); k-means warm start sampling: pick a component, draw from it loss: mixture negative log-likelihood density: exact $\log p$ 92,906 parameters	cVAE: conditional VAE head: posterior $q(z \mathbf{a}, e)$, decoder $p(\mathbf{a} z, e')$ (Gaussian, learned variance); latent z: 8 sampling: $z \sim \mathcal{N}(0, I)$, decode loss: reconstruction NLL + β KL ($\beta = 0.1$, warm-up over 30 epochs) density: $\log p$ by importance sampling 148,128 parameters	Flow: conditional flow head: 8 affine coupling layers conditioned on e (masks alternate, 96 hidden units) sampling: $z \sim \mathcal{N}(0, I)$ through the inverse loss: exact negative log-likelihood density: exact $\log p$ 209,600 parameters
Diffusion: conditional diffusion head: MLP denoiser (128) of noisy design, step and e sampling: 100 reverse steps from noise loss: MSE of the predicted noise density: none 93,640 parameters	BasisFlow: basis + flow head: spectrum → Gaussian over 8 coefficients (learned basis); coupling flow (8 layers) maps coefficients ↔ design loss: NLL + 0.1 basis reconstruction + 0.1 forward consistency; exact $\log p$ 206,845 parameters	PadINN: padded invertible net head: one bijection between the zero-padded design and [spectrum; z], z: 16 numbers sampling: invert $[y, z]$, $z \sim \mathcal{N}(0, I)$ loss: spectrum MSE + design MSE + $50 \times \text{MMD}(z)$; density: none 810,704 parameters

Training and evaluation



Design $\mathbf{a}$: the two resonators sorted by f_t , each $[m, f_t, x, y]$, mapped by a logit to unbounded coordinates (bounded design space; $k = m(2\pi f_t)^2$).

Data: 100k configurations, 2 resonators on a 14×5 position grid, 18 plate modes, 80/10/10 split. Spectrum encoder: shared by MDN, cVAE (two copies), Flow, Diffusion.